RIT | Rochester Institute of Technology of Dubai

RIT Dubai 16th Engineering Competition

Academic Year 2026 – 2027

Software Engineering Competition

Two-Phases (prepared and 2-hours onsite) Competition

Phase 2 Rapid Integration & Operational Surprises (On competition Day)

Task: Nearest Medical Center Dispatch & On-Call Doctor Assignment

Objectives:

Extend the system to match medical requests from shelters (e.g., "Requires On-Duty Doctor", "Pediatrician Needed", or "Trauma Specialist") with available medical personnel at nearby hospitals or medical centers based on minimum distance and doctor specialization.

Challenge Title	Emergency Relief Operations & Dispatch System
Target Audience	High School Students (Grades 9–12)
Phase 2 (Competition Day)	2 Hours Live Hackathon | Implementing new added features and dataset
Team & Submission	Max 4 Students per Team | GitHub Repo / ZIP Source Code + Live Demo Pitch

Detailed Description:

In addition to physical supplies, emergency shelters often require immediate medical assistance. Teams receive a new dataset containing **Hospitals/Medical Centers** (ID, name, location coordinates (x, y), list of available doctor IDs, and specialties) and **On-Call Doctors** (ID, specialty, availability status).

When a shelter logs a medical request, the system must:

1. **Specialization & Availability Filter:** Identify hospitals that currently have an *available* doctor matching the requested specialty (e.g., General Practitioner, Triage Doctor, Pediatrician).
2. **Nearest Hospital Routing:** Calculate the Euclidean distance $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ between the shelter and candidate hospitals, selecting the **nearest facility** with an eligible doctor.
3. **Status Update & Allocation:**
    - Assign the doctor to the shelter request ticket.
    - Automatically toggle the assigned doctor's status from Available to Dispatched/Busy to prevent double-booking.
4. **Fallback Handling (Edge Case):** If no matching specialist is available at the nearest hospital, fall back to the next-closest hospital or mark the ticket as "Pending Medical Dispatch - High Urgency".
5. **Dashboard & Report Update:** Display the assigned Doctor ID, Specialty, Hospital Name, and Travel Distance on the administrative UI dashboard and include these details in the exported CSV/JSON reports.

Deliverables

- A ZIP file containing clean source code for the whole system, installation README.md, link to *GitHub* repository.
- Phase 2: A 3-minute video presentation/demo for all features implemented during competition, integration process and what new features you can add to improve the system.

Permitted Tools & AI Policy

- **Allowed Tech Stack:** Python, JavaScript/Node.js, HTML/CSS/JS (Vanilla or React), Tkinter, Flask, or Pandas.
- **AI Tooling Policy:** Generative AI tools (e.g., ChatGPT, Copilot) are restricted to code debugging and syntax lookup. Generating complete codebase solutions using AI is strictly prohibited.